

Correlation

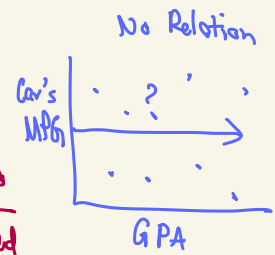
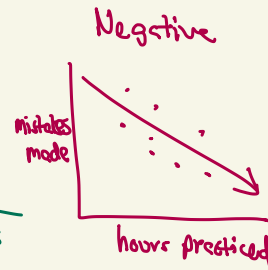
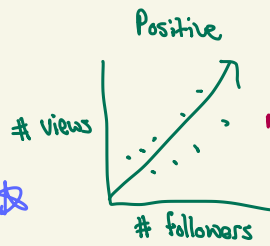
- Relationship between 2 variables

→ how strong?

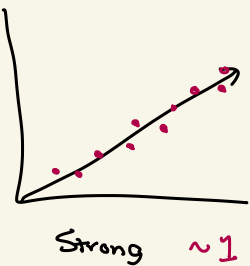
→ the direction

Directions

- Positive trends ★
- Negative trends ✖
- No trends (no relationship) ✖



Strength



Covariance

- To check whether 2 variables are related, we ask do they **Covary**?
- We answer with **Covariance**
 - how much each value deviates from mean
 - if both variables deviate in the same way, they are likely related

Standard Deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Variance

$$v = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Covariance

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

- Spread of data from mean
- $\bar{x}$ = mean
- x_i = individual values

- $\text{Cov}(x, y)$ is similar to Variance, but we multiply by difference of y instead of squaring x

Data

Viewers	1	2	3	4	5
ads seen x	5	4	4	6	8
items purchased y	8	9	10	13	15

$$\bar{x} = 5.4$$

$$\sigma_x = 1.07$$

$$\bar{y} = 11$$

$$\sigma_y = 2.92$$

Calculating time! :-

x_i	$x_i - \bar{x}$
5	-0.4
4	-1.4
4	-1.4
6	0.6
8	2.6

y_i	$y_i - \bar{y}$
8	-3
9	-2
10	-1
13	2
15	4

$(x_i - \bar{x}) \cdot (y_i - \bar{y})$
1.2
2.8
1.4
1.2
10.4

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 17$$

$$\text{Cov}(x, y) = \frac{17}{n-1} = \frac{17}{5-1} = \frac{17}{4} = 4.25$$

What does this tell us?

- ⊕ positive indicates as one variable deviates, the other deviates in the **same** way
- ⊖ Negative indicates as one variable deviates, the other deviates in the **opposite** way

Is covariance correlation? **NO.**

- Correlation can sometimes be measured with cov

e.g., $\left. \begin{array}{l} \text{Cov in } ^\circ\text{F} = 10 \\ \text{Cov in } ^\circ\text{C} = 60 \end{array} \right\} \text{Same data, different units}$

Covariance is NOT standardized.

→ we would have to convert all data to the same unit

Correlation Coefficient (r)

* Also
"Pearson Correlation Coefficient"
Python → `pearsonr()`

- This is standardized cov

→ will always be between -1 and 1

→ -1 indicates perfect negative relationship

→ 1 indicates perfect positive relationship

→ 0 indicates no linear relationship

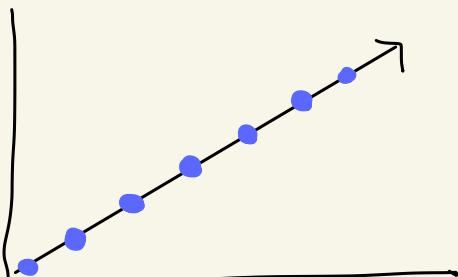
↑
direction

$r = \pm 1$	→ Perfect
$r = \pm 0.8 - 0.9$	→ Strong
$r = \pm 0.5 - 0.7$	→ Medium
$r = \pm 0.1 - 0.4$	→ Weak
$r = 0$	→ no relationship

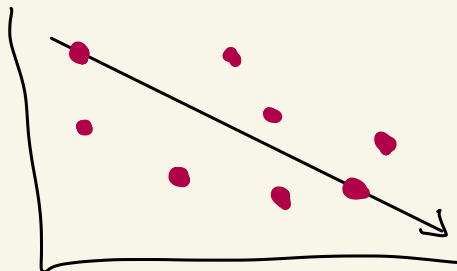
↑
Strength

$$r = \frac{\text{cov}(x, y)}{\sigma_x \cdot \sigma_y} = \frac{4.25}{(1.67)(2.92)} = 0.87$$

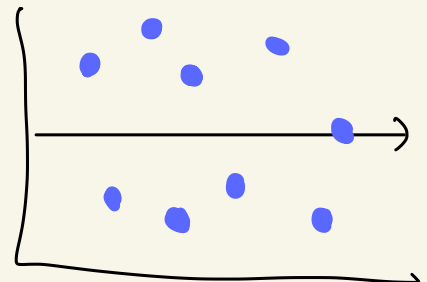
σ → standard dev.



$r = 1$



$r = -0.4$



$r = 0$